Instrumentation Lab Session 2 Sensors and Transducers

1. Objectives

In this lab session we will look at some input and output transducers and build some 'conditioning' electronics to filter and amplify signals. We'll use sound (air pressure) transducers since, as well as having some interesting properties in their own right, these allow us to work with a quantity which we can both hear and measure electronically.

The input transducer will be an Electret Microphone. This is a capacitive-type input transducer; the incident sound pressure waves cause one flexible side of a capacitor to vibrate; this movement modifies the capacitance hence this can be picked-up by applying a voltage to the device. Capacitive microphones normally require a relatively high voltage to operate well (usually 50V), however the Electret type has a capacitor which is 'ready-charged' by a static charge embedded within the polymer plates of the capacitor. It also incorporates a built-in FET transistor to match the extremely high impedance of the Electret capacitance and provide some signal amplification.

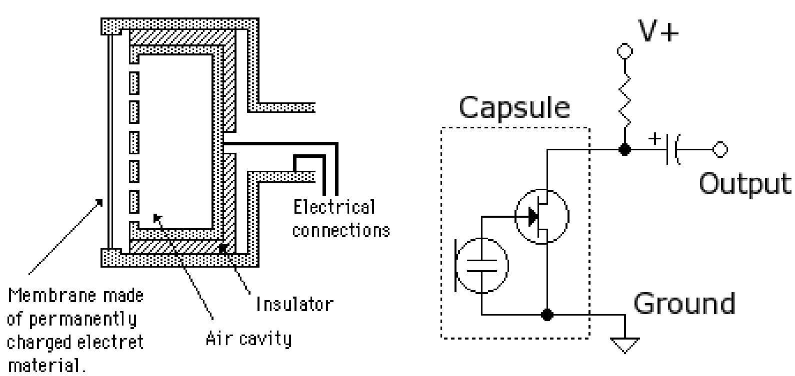


Figure 1-1, Electret Microphone Schematic

The output transducer will be a Piezo-electric Sounder. This is a diaphragm of piezo-electric material to which an AC voltage signal is applied. Through the piezo-electric effect, the applied voltage results in mechanical deformation, hence generation of sound waves.

Figure 1-2, Piezo-Electric Sounder

The sounder functions as a crude loudspeaker, though it has a many resonances as we'll see later. The device we'll use is basically similar to the picture above, inside a plastic case.

In this lab we will build a sound pressure transducer (**microphone**) circuit and investigate its properties. To do this we will need to build an **amplifier** (using an **op-amp)** for the microphone and a **filter** to select only the frequency regions we are interested in.

2. Technical Data for Our Input / Output Transducers

When purchasing a sensor, the manufacturer would normally supply you with a comprehensive datasheet that would state all the parameters you need to know in order to use the device as a measurement sensor. These would be parameters such as sensitivity, bandwidth, frequency response curve etc. Our microphone is not a measurement device (its datasheet is far from being comprehensive). In the absence of a datasheet, the sensor could be sent to a qualified lab for calibration and the lab would be able to work out what the specs of your sensor are. Sensor and instrument calibration is a complicated and profitable business. There are legal requirements for periodic calibration and certification of measurement instruments used in commercial or medical environments and critical manufacturing processes such as drug and food production plant. In those cases, knowing your sensor is vital!
However, our microphone is of rather inferior quality. There are two reasons for this.

1. Using a poor quality device helps us to see better the kind of real-world effects which we need to understand
2. We've already spent a lot of money on the ELVIS units you're sitting in front of

Technical data for the 2 transducers can be found in the Appendix.

3. Build the microphone

- Power off the prototyping board.
- Wire up the Electret microphone on the prototyping board according to Figure 3-1 and Figure 3-2. Figure 3-2 gives you an indication of where to place the circuit on the board (you will use the rest of board to build other circuits) and what it should look like. Your circuit needn't be a replica of that in Figure 3-2, however **remember to keep the circuit tidy by having short wires and routing them logically**
- Use a 1kΩ resistor in-line with the +15V supply to protect the device (it will limit the current drawn by the device). It also sets the output impedance.
- Note: make sure you connect to ground the correct leg of the microphone capsule i.e. the one that's connected with the capsule's can. **If not sure which one it is, use the DMM to measure continuity between a leg and the can.**

3.1 DC-Coupled Output

- Use either the Tektronix or ELVIS scope to observe the microphone output waveform available between GND and point (2) in the circuit. Make use of the BNC connectors on the prototyping board. **Make sure you set the scope input coupling to DC.**
- Provide suitable input stimulation! (i.e. make some noises). How would you describe the signal? What are the problems associated with this kind of signal and how can you overcome this? Discuss with your demonstrator.

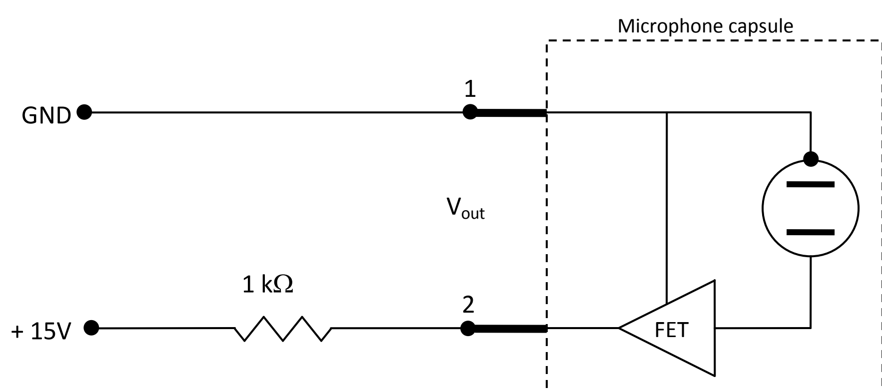


Figure 3-1, Electret Microphone – Circuit Diagarm

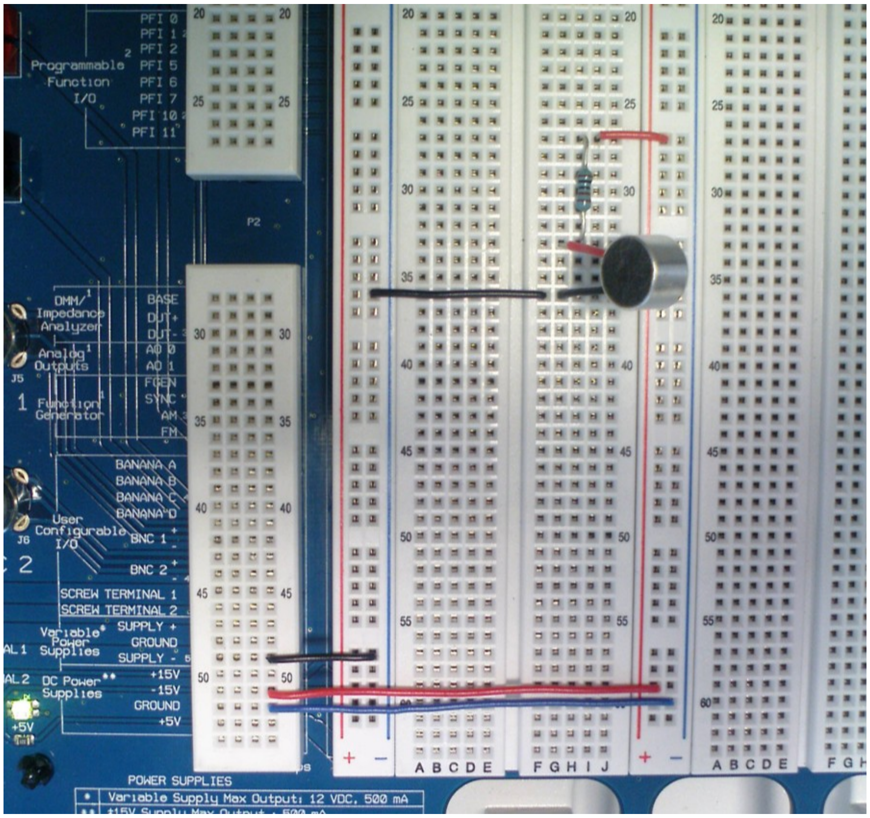


Figure 3-2, Electret Microphone – Circuit Implementation

3.2 AC Coupled Output

- Sound is of course represented by an AC signal. The easiest way to decouple the AC from the DC component of the signal is to use a suitably large blocking capacitor.
- Modify the circuit according to Figure 3-3 and Figure 3-4. The higher frequencies will be available across the resistor. You are using the resistor and capacitor in series as a high-pass filter.
- What factors dictate the choice of C and R values? First, consider what should be the bandwidth of your microphone and the limitations imposed by the Electret capsule's frequency response (see section 5.1). What is the lowest frequency you want to pass?
- Calculate the 3dB of the filter you need and hence its time constant. What value of C and R do you wish to use, and can you find such in the component stock? If not, you may have to compromise and use a different value. In any case, note down the cut-off frequency of the filter you have built.

  NOTE: avoid extreme resistor values: choose a resistor between 10 $k\Omega$ and 1 $M\Omega$.

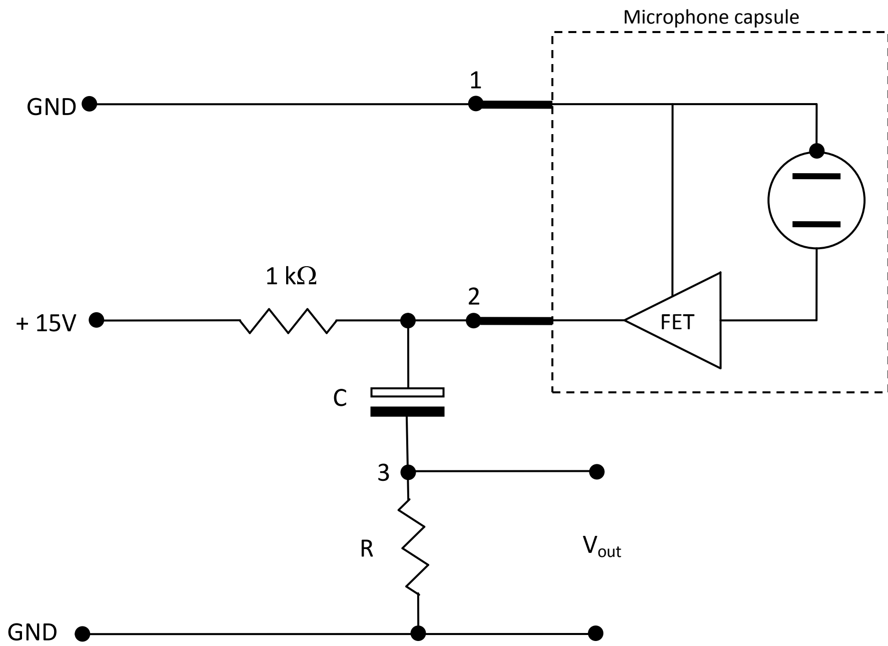


Figure 3-3, Electret Microphone with Output Filter - Circuit Diagarm

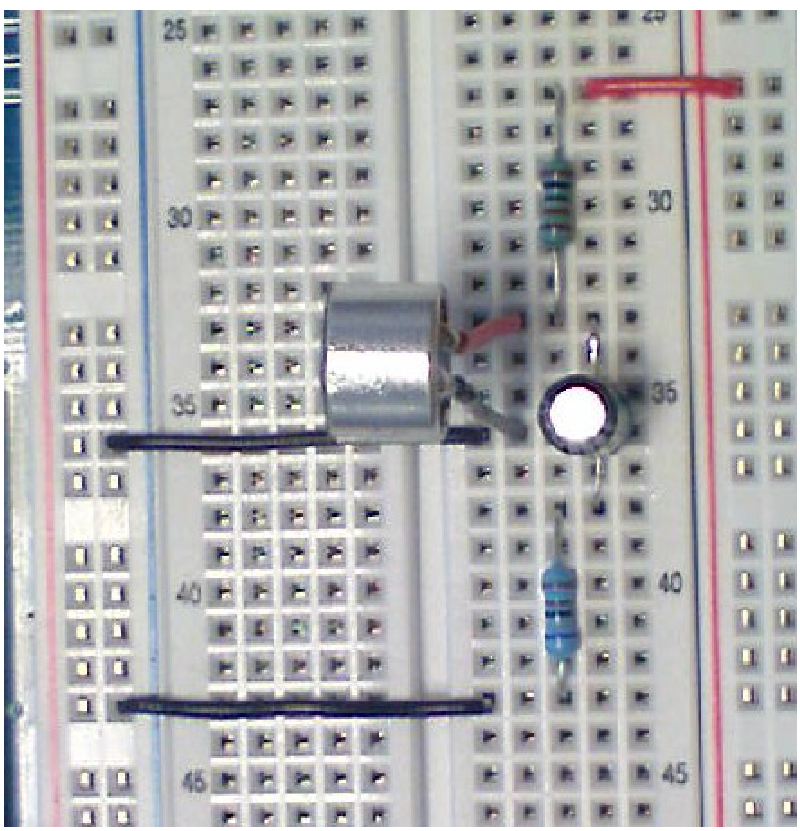


Figure 3-4, Electret Microphone with Output Filter - Circuit Implementation

- Connect filter output available between GND and (3) to the scope. Make use of the BNC connectors on the prototyping board as appropriate.
- Re-test the microphone (shout at it, or provide some other input)

To further test it, we'll use the sounder, which generates a more controlled signal

- Place the sounder in front of the microphone.
- Wire it to FGEN
- Set FGEN to sine wave, 2.7 kHz, V_{pp} = 0.40V
- Switch prototyping board and FGEN on

You may find that the sounder is rather loud and uncomfortable. Establish a firm coupling between the sounder and the microphone using the plastic tube and the rubber band as shown in Figure 3-5. This will help keep the noise down.

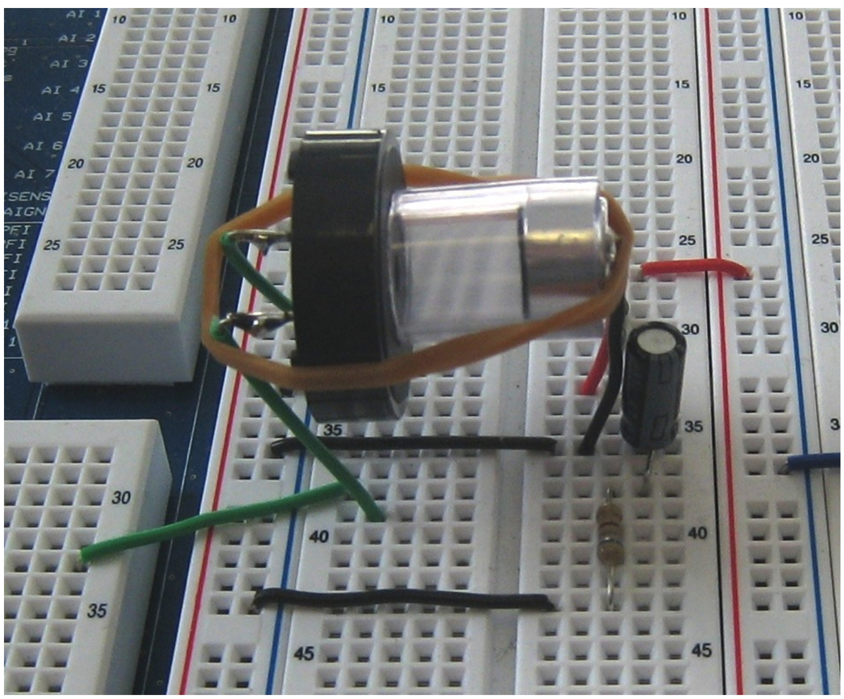


Figure 3-5, Reducing the Noise from the Sounder!

For the same purpose, you may want to switch FGEN off when not measuring. The observed V_{out} may change slightly as a consequence of the presence of the coupler.

- Observe the value of the microphone's output; it should be in the order of mV. Also, the signal is rather noisy. The Signal to Noise Ratio (SNR) is rather poor.
- Try increasing the amplitude of the FGEN a bit (but not too much – keep the noise down to an acceptable limit).

- At a tolerable volume, make a note of the FGEN driving amplitude.
- Run ELVIS's DSA (Dynamic Signal Analyser). The DSA is in essence an FFT analyser - similar to the program FFTAnalyser.vi you used in session 1 - in that it provides information about the frequency content of a signal but instead of measuring the amplitude spectrum of a signal, it measures its power spectrum. In the latter, the amplitudes of the frequency lines are scaled to represent the "power" content of the signal. See the Appendix for more information on amplitude spectrum and power spectrum.
- Note that the 'FFT Settings' on the DSA settings are different from those of FFTAnalyser.vi. In the latter you had "number of samples [S]" and "sampling rate [S/s]"; here you have "frequency span [Hz]" and "resolution [lines]".
- Look at Help to understand the meaning of the settings and in particular how "number of samples [S]" and "sampling rate [S/s]" relate to "frequency span [Hz]" and "resolution [lines]". (Don't spend too much time on this, ask a demonstrator if you need to).
- Use the power-spectrum measurement capabilities of ELVIS to make a rough measurement of the SNR (Signal-to-Noise Ratio, use the definition given in the Appendix)

ELVIS digitiser has a much larger input range (±10V) than we are making use of at present. To make the most out of the dynamic range of this measurement device, we need amplify the microphone's output.

4. Build the Preamplifier

4.1 LM741 Op-Amp

- Place a LM741 op-amp into the breadboard and power it up. Use your LM741 datasheet to work out the pins. What is the output voltage and why?

4.2 Inverting Preamplifier Circuit

- Following Figure 4-1 and Figure 4-2, build an inverting preamplifier using the LM741 op-amp. Build this separately from the microphone circuit; we'll wire them both together later.

Note: this circuit is called a 'preamplifier' and it is intended to boost a signal's voltage level only. A 'power amplifier' on the other hand, is used to drive a heavy load.

- Choose the values of R_1 and R_f to give a gain A of approximately 10. Before you insert the resistors into the circuit, measure their actual values using the DMM.

NOTE: use values of R_1 and $R_f \geq 5k\Omega$.

- Make some basic checks to test the functionality of the amplifier.

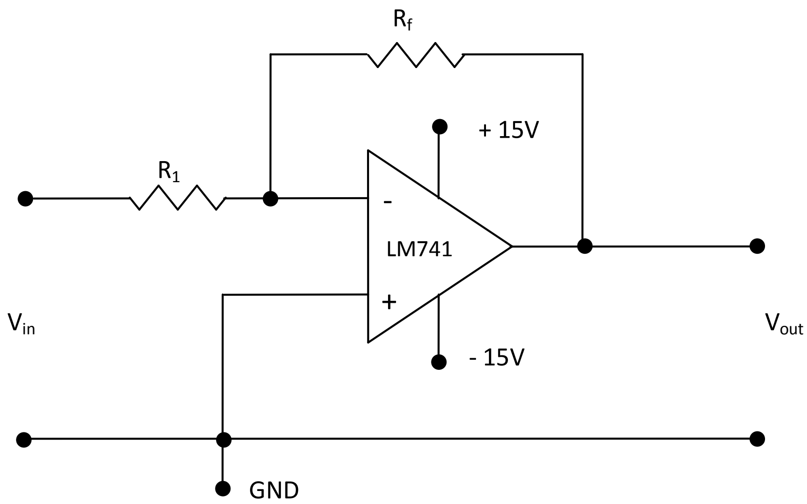


Figure 4-1, Pre-Amplifier – Circuit Diagram

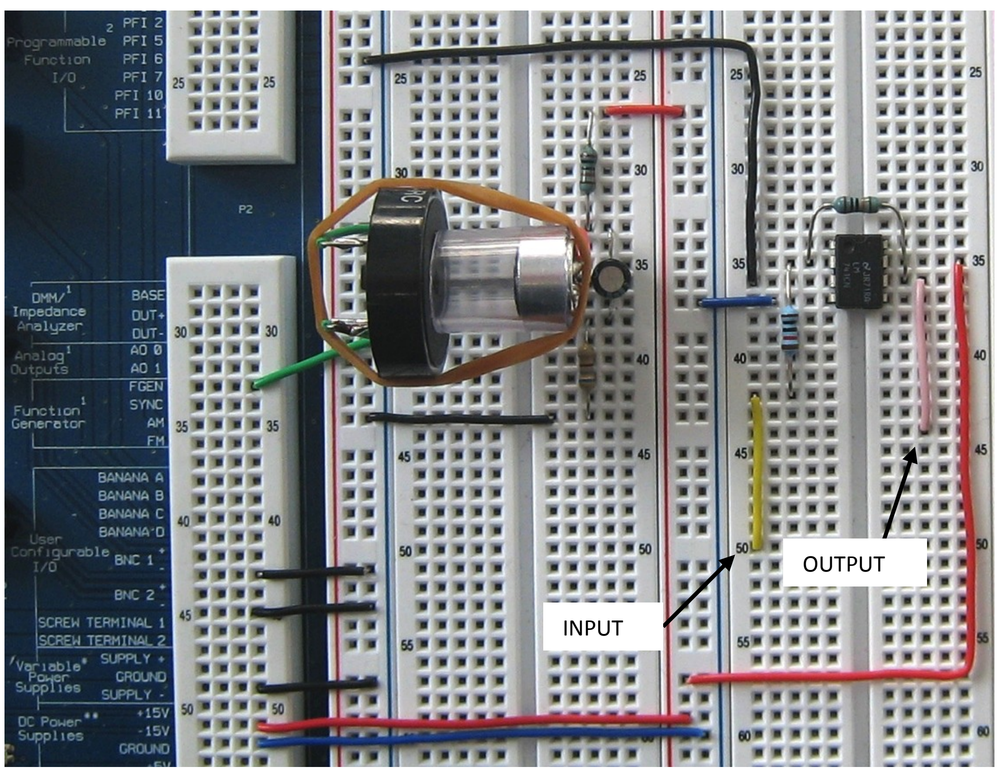


Figure 4-2, Pre-Amplifier – Circuit Implementation

4.3 Measure gain for different DC voltages

Use the ELVIS VPS (Variable Power Supply) to provide V_{in} (read Help to find out how the VPS instrument works).

*NOTE: Make sure you use the VPS **negative voltage line**. We have found that the positive line exhibits a variable offset which will compromise the linearity measurement!*

- Use the ELVIS DMM to read V_{out}. Make use of the "banana" connectors available on the prototyping board
- Ramp up V_{in} in a number of steps to cover the whole output range of the pre-amp.
- Check pre-amp linearity by plotting V_{out}/V_{in} graph. Use the definition given in the lectures to make a rough assessment of the amplifier linearity.
- What's the actual gain? How does this compare with the value expected, based on the actual resistor values?
- What is the amplifier DC offset (V_{out} for zero V_{in})?

4.4 Testing with periodic signals

- Use FGEN to provide V_{in}; connect V_{in} to Tektronix CH1 and display it.
- Set FGEN to output a sine wave, 0.2 V_{pp}, 80Hz
- Connect V_{out} to the DMM and Tektronix CH2; display the waveform. Make use of the BNC and "banana" connectors available on the prototyping board as you see fit.
- Set the DMM to measure a periodic (AC) voltage
- Observe the phase difference between V_{in} and V_{out}
- Ramp the frequency up to 200 kHz in suitable steps and see what happens to the pre-amp gain and V_{out} phase (with respect to pre-amp V_{in} phase).

Note: You can use FGEN in "Manual Mode" by ticking the Manual checkbox and using the knob on the ELVIS to vary the frequency manually.

- Look at the shape of V_{out} at 200kHz. Would you expect this?
- What is the bandwidth? (determine the 3dB point)
- Measure the Gain Cross-over frequency and the Phase Cross-over frequency. Use the definitions given in lectures (and also repeated in the Appendix). Note: you may need to observe signals with frequencies beyond the capability of the ELVIS unit for this; try using the Tektronix scope.

5. Appendices

5.1 Electret Microphone Capsule Technical Specifications

Manufacturer:	unknown (low-cost part)
Response:	Omni-directional
Bandwidth:	50Hz to 13kHz
Sensitivity:	-60dB ±3dB (0dB = 1V/µbar at 1kHz)
Impedance:	1kΩ maximum
S/N ratio:	>40dB
Sound pressure level:	120dB maximum
Power supply:	10 to 15V DC
Current drain:	<4mA

5.2 Piezo-Electric Sounder Technical Specifications

Rated voltage	$9V_{pp}$ (Square Wave)
Operating voltage:	$1\text{-}30V_{pp}$
Rated current at rated voltage:	3mA
Capacitance at 120Hz:	17000pF (±30%)
Sound output at 4000Hz at 10cm at rated voltage:	>85dB
Resonant frequency:	4000 ±500 Hz
Operating temperature:	-20 ~ +60°C
Storage temperature:	-30 ~ +70°C
Weight:	3g

5.3 Signal to Noise Ratio

$$SNR = \frac{P_{signal}}{P_{noise}}$$

Often expressed in dB: $SNR = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$

5.4 Amplifier Stability

Gain cross-over is the frequency where Gain = 1 (or 0dB)

Phase cross-over is the frequency where phase = -180°

Gain margin is the difference in gain at these two frequencies

Phase margin is the difference in phase at these two frequencies

The amplifier is stable if we have both a positive (non-zero) gain and phase margin.

In practical systems, we like to have a gain margin > 6dB and a phase margin of more than 45°

5.5 Notes on amplitude spectrum and power spectrum

An FFT Analyser measures the Amplitude Spectrum of a signal. The spectrum is measured on one block of data and no averaging is performed. The spectrum consists of a number of components (one per frequency), which are most often displayed as magnitude and phase information. A DSA measures the Power Spectrum (also known as Autospectrum, Autopower Spectrum, or just Spectrum). These terms refer to an average of the power of the individual frequency components over a number of instantaneous spectra. When this averaging occurs, the magnitude of each frequency component is squared, and added to the previous sum for that frequency. Hence, the phase information is discarded. The averaging to compute a power spectrum does not reduce the noise (unlike time-domain averaging to reduce unwanted noise). Hence, it can be extremely useful in getting a reliable statistical estimate of the level of random signals being measured. For example, if you only display the instantaneous spectrum of white noise, the spectrum will be extremely 'jagged'. But after computing the power spectrum with adequate averaging, the shape of the spectrum will converge on a flat spectrum. Power spectra can be shown in units of power, or the square root of the averaged result can be taken to give a voltage value (which is the RMS value).

In addition, the power spectrum may be scaled in different ways, depending on the type of signal you are analysing, for instance Power Spectral Density (PSD) and Energy Spectral Density (ESD). PSD is used when measuring continuous broadband noise, and normalizes the power to an equivalent bandwidth of 1 Hz, irrespective of the actual bandwidth of the filter being used. For example, if a signal is measured at -93 dB in a 10 Hz bandwidth, then the spectral density would be -103 dB (in a 1 Hz bandwidth). This makes it possible to compare noise measurements made with different bandwidth settings of the spectrum analyser.

5.6 LM741 data sheet excerpt

National Semiconductor

August 2000

LM741
Operational Amplifier

General Description

The LM741 series are general purpose operational amplifiers which feature improved performance over industry standards like the LM709. They are direct, plug-in replacements for the 709C, LM201, MC1439 and 748 in most applications.

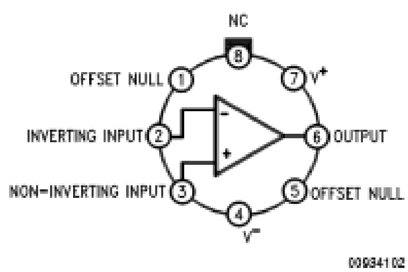
The amplifiers offer many features which make their application nearly foolproof: overload protection on the input and output, no latch-up when the common mode range is exceeded, as well as freedom from oscillations.

The LM741C is identical to the LM741/LM741A except that the LM741C has their performance guaranteed over a 0°C to +70°C temperature range, instead of −55°C to +125°C.

Features

Connection Diagrams

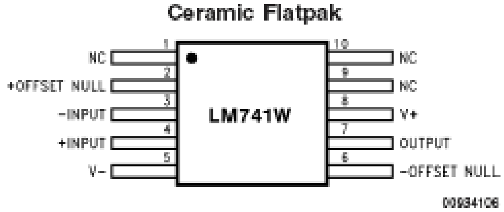
Metal Can Package



Note 1: LM741H is available per JM38510/10101

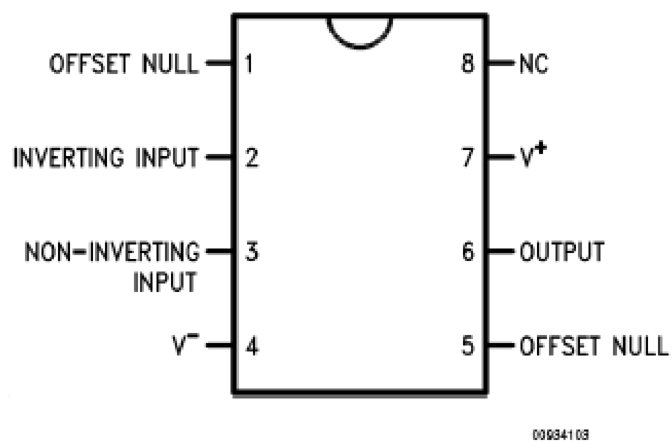
Order Number LM741H, LM741H/883 (Note 1), LM741AH/883 or LM741CH
See NS Package Number H08C

Ceramic Flatpak



Order Number LM741W/883
See NS Package Number W10A

Dual-In-Line or S.O. Package



Order Number LM741J, LM741J/883, LM741CN
See NS Package Number J08A, M08A or N08E

Absolute Maximum Ratings (Note 2)

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/ Distributors for availability and specifications.

(Note 7)

	LM741A	LM741	LM741C
Supply Voltage	±22V	±22V	±18V
Power Dissipation (Note 3)	500 mW	500 mW	500 mW
Differential Input Voltage	±30V	±30V	±30V
Input Voltage (Note 4)	±15V	±15V	±15V
Output Short Circuit Duration	Continuous	Continuous	Continuous
Operating Temperature Range	−55°C to +125°C	−55°C to +125°C	0°C to +70°C
Storage Temperature Range	−65°C to +150°C	−65°C to +150°C	−65°C to +150°C
Junction Temperature	150°C	150°C	100°C
Soldering Information			
N-Package (10 seconds)	260°C	260°C	260°C
J- or H-Package (10 seconds)	300°C	300°C	300°C
M-Package			
Vapor Phase (60 seconds)	215°C	215°C	215°C
Infrared (15 seconds)	215°C	215°C	215°C

See AN-450 "Surface Mounting Methods and Their Effect on Product Reliability" for other methods of soldering
surface mount devices.

	LM741A	LM741	LM741C
ESD Tolerance (Note 8)	400V	400V	400V

Electrical Characteristics (Note 5)

Parameter	Conditions	LM741A			LM741			LM741C			Units
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage	$T_A = 25°C$										
	$R_S \leq 10\ k\Omega$					1.0	5.0		2.0	6.0	mV
	$R_S \leq 50\Omega$		0.8	3.0							mV
	$T_{AMIN} \leq T_A \leq T_{AMAX}$										
	$R_S \leq 50\Omega$			4.0							mV
	$R_S \leq 10\ k\Omega$						6.0			7.5	mV
Average Input Offset Voltage Drift				15							μV/°C
Input Offset Voltage Adjustment Range	$T_A = 25°C$, $V_S = \pm 20V$	±10				±15			±15		mV
Input Offset Current	$T_A = 25°C$		3.0	30		20	200		20	200	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$			70		85	500			300	nA
Average Input Offset Current Drift				0.5							nA/°C
Input Bias Current	$T_A = 25°C$		30	80		80	500		80	500	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$			0.210			1.5			0.8	μA
Input Resistance	$T_A = 25°C$, $V_S = \pm 20V$	1.0	6.0		0.3	2.0		0.3	2.0		MΩ
	$T_{AMIN} \leq T_A \leq T_{AMAX}$, $V_S = \pm 20V$	0.5									MΩ
Input Voltage Range	$T_A = 25°C$							±12	±13		V
	$T_{AMIN} \leq T_A \leq T_{AMAX}$				±12	±13					V

Electrical Characteristics (Note 5) (Continued)

Parameter	Conditions	LM741A			LM741			LM741C			Units
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Large Signal Voltage Gain	$T_A = 25°C$, $R_L \geq 2\ k\Omega$										
	$V_S = \pm20V$, $V_O = \pm15V$	50									V/mV
	$V_S = \pm15V$, $V_O = \pm10V$				50	200		20	200		V/mV
	$T_{AMIN} \leq T_A \leq T_{AMAX}$,										
	$R_L \geq 2\ k\Omega$,										
	$V_S = \pm20V$, $V_O = \pm15V$	32									V/mV
	$V_S = \pm15V$, $V_O = \pm10V$				25			15			V/mV
	$V_S = \pm5V$, $V_O = \pm2V$	10									V/mV
Output Voltage Swing	$V_S = \pm20V$										
	$R_L \geq 10\ k\Omega$	±16									V
	$R_L \geq 2\ k\Omega$	±15									V
	$V_S = \pm15V$										
	$R_L \geq 10\ k\Omega$				±12	±14		±12	±14		V
	$R_L \geq 2\ k\Omega$				±10	±13		±10	±13		V
Output Short Circuit	$T_A = 25°C$	10	25	35		25			25		mA
Current	$T_{AMIN} \leq T_A \leq T_{AMAX}$	10		40							mA
Common-Mode	$T_{AMIN} \leq T_A \leq T_{AMAX}$										
Rejection Ratio	$R_S \leq 10\ k\Omega$, $V_{CM} = \pm12V$				70	90		70	90		dB
	$R_S \leq 50\Omega$, $V_{CM} = \pm12V$	80	95								dB
Supply Voltage Rejection	$T_{AMIN} \leq T_A \leq T_{AMAX}$,										
Ratio	$V_S = \pm20V$ to $V_S = \pm5V$										
	$R_S \leq 50\Omega$	86	96								dB
	$R_S \leq 10\ k\Omega$				77	96		77	96		dB
Transient Response	$T_A = 25°C$, Unity Gain										
Rise Time			0.25	0.8		0.3			0.3		μs
Overshoot			6.0	20		5			5		%
Bandwidth (Note 6)	$T_A = 25°C$	0.437	1.5								MHz
Slew Rate	$T_A = 25°C$, Unity Gain	0.3	0.7			0.5			0.5		V/μs
Supply Current	$T_A = 25°C$					1.7	2.8		1.7	2.8	mA
Power Consumption	$T_A = 25°C$										
	$V_S = \pm20V$		80	150							mW
	$V_S = \pm15V$					50	85		50	85	mW
LM741A	$V_S = \pm20V$										
	$T_A = T_{AMIN}$			165							mW
	$T_A = T_{AMAX}$			135							mW
LM741	$V_S = \pm15V$										
	$T_A = T_{AMIN}$					60	100				mW
	$T_A = T_{AMAX}$					45	75				mW

Note 2: "Absolute Maximum Ratings" indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is functional, but do not guarantee specific performance limits.